Input Context

```
Write a high-quality answer for the given question using only the provided search
results (some of which might be irrelevant).

Document [1](Title: Asian Americans in science and technology) Prize in physics for
discovery of the subatomic particle J/ψ. Subrahmanyan Chandrasekhar shared...
Document [2](Title: List of Nobel laureates in Physics) The first Nobel Prize in
Physics was awarded in 1901 to Wilhelm Conrad Röntgen, of Germany, who received...
Document [3](Title: Scientist) and pursued through a unique method, was essentially
in place. Ramón y Cajal won the Nobel Prize in 1906 for his remarkable...

Question: who got the first nobel prize in physics
Answer:
```

Desired Answer

```
Wilhelm Conrad Röntgen
```

Figure 2: Example of the multi-document question answering task, with an input context and the desired model answer. The document containing the answer is bolded within the input context here for clarity.

Input Context

```
Write a high-quality answer for the given question
using only the provided search results (some of
which might be irrelevant).

Document [1](Title: List of Nobel laureates in
Physics) ...
Document [2](Title: Asian Americans in science and
technology) ...
Document [3](Title: Scientist) ...

Question: who got the first nobel prize in physics
Answer:
```

Desired Answer

```
Wilhelm Conrad Röntgen
```

Figure 3: Modulating the position of relevant information within the input context for the multi-document question answering example presented in Figure 2. Reordering the documents in the input context does not affect the desired output.

Input Context

```
Write a high-quality answer for the given question
using only the provided search results (some of
which might be irrelevant).

Document [1](Title: Asian Americans in science and
technology) ...
Document [2](Title: List of Nobel laureates in
Physics) ...
Document [3](Title: Scientist) ...
Document [4](Title: Norwegian Americans) ...
Document [5](Title: Maria Goeppert Mayer) ...

Question: who got the first nobel prize in physics
Answer:
```

Desired Answer

```
Wilhelm Conrad Röntgen
```

Figure 4: Modulating the input context length of the multi-document question answering example presented in Figure 2. Adding documents that do not contain the answer increases the length of the input context, but does not affect the desired output.

Our experimental setup is similar to the needle-in-a-haystack experiments of Ivgi et al. (2023), who compare question answering performance when the relevant paragraph is placed (i) at the beginning of the input or (ii) a random position within the input. They find that encoder-decoder models have significantly higher performance when relevant information is placed at the start of the input context. In contrast, we study finer-grained changes in the position of relevant information.

2.2 Models

We analyze several state-of-the-art open and closed language models. We use greedy decoding when generating outputs and leave exploration of other decoding methods to future work. We use a standard set of prompts for each model (Figure 2).

Open models. We experiment with MPT-30B-Instruct, which has a maximum context length of 8192 tokens. The model was initially pre-trained on 1 trillion tokens using 2048-token sequences, followed by an additional sequence length adaptation pre-training phase on 50 billion tokens using 8192-token sequences. MPT-30B-Instruct uses ALiBi (Press et al., 2022) to represent positional information. We also evaluate LongChat-13B (16K) (Li et al., 2023), which extends the LLaMA-13B (Touvron et al., 2023a) context window from 2048 to 16384 tokens by using condensed rotary positional embeddings before fine-tuning with 16384-token sequences.

Closed models. We use the OpenAI API to experiment with GPT-3.5-Turbo and GPT-3.5-Turbo

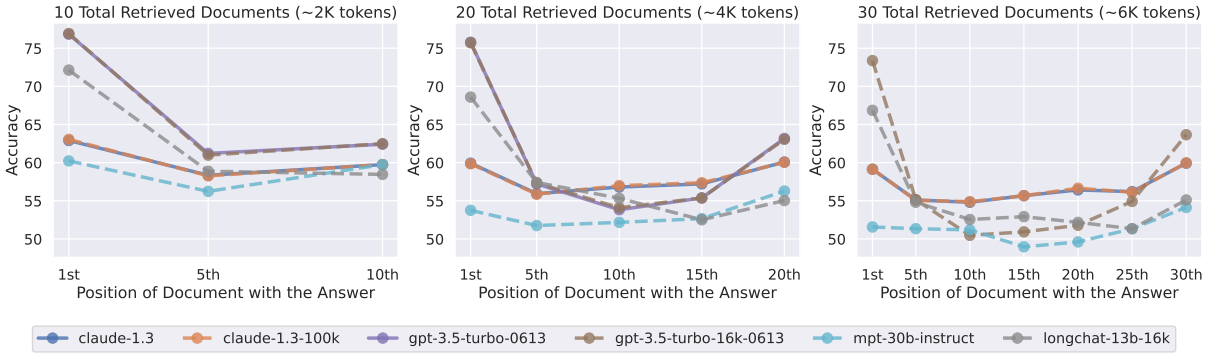

Figure 5: The effect of changing the position of relevant information (document containing the answer) on multi-document question answering performance. Lower positions are closer to the start of the input context. Performance is highest when relevant information occurs at the very start or end of the context, and rapidly degrades when models must reason over information in the middle of their input context.

(16K).[5] GPT-3.5-Turbo has a maximum context length of 4K tokens, and GPT-3.5-Turbo (16K) is a version with an extended maximum context length of 16K tokens. We evaluate Claude-1.3 and Claude-1.3 (100K) with the Anthropic API; Claude-1.3 has a maximum context length of 8K tokens, and Claude-1.3 (100K) has an extended context length of 100K tokens. [6]

2.3 Results and Discussion

We experiment with input contexts containing 10, 20, and 30 total documents. Figure 5 presents multi-document question answering performance when varying the position of relevant information within the input context. To contextualize model performance, we also evaluate on the closed-book and oracle settings (Table 1). In the closed-book setting, models are not given any documents in their input context, and must rely on their parametric memory to generate the correct answer. On the other hand, in the oracle setting, language models are given the single document that contains the answer and must use it to answer the question.

Model performance is highest when relevant information occurs at the beginning or end of its input context. As illustrated in Figure 5, changing the position of relevant information in the input context leads to substantial decreases in model performance. In particular, we see a distinctive U-shaped performance curve—models are often much better at using relevant information that occurs at the very beginning (primacy bias) and very end of contexts (recency bias), and suffer degraded performance when forced to use information within the middle of its input context. For example, GPT-3.5-Turbo's multi-document QA performance can drop by more than 20%—in the worst case, performance in 20- and 30-document settings is lower than performance without *any* input documents (i.e., closed-book performance; 56.1%). These results indicate that current models cannot effectively reason over their entire context window when prompted for downstream tasks.

Model	Closed-Book	Oracle
LongChat-13B (16K)	35.0%	83.4%
MPT-30B-Instruct	31.5%	81.9%
GPT-3.5-Turbo	56.1%	88.3%
GPT-3.5-Turbo (16K)	56.0%	88.6%
Claude-1.3	48.3%	76.1%
Claude-1.3 (100K)	48.2%	76.4%

Table 1: Closed-book and oracle accuracy of language models on the multi-document question answering task.

Extended-context models are not necessarily better at using input context. When the input context fits in the context window of both a model and its extended-context counterpart, we see that performance between them is nearly identical. For example, the 10- and 20-document settings both fit in the context window of GPT-3.5-Turbo and GPT-3.5-Turbo (16K), and we observe that their performance as a function of position of relative information is nearly superimposed (solid purple and dashed brown series in Figure 5). These results

[5] We use the `0613` OpenAI model versions.

[6] We also evaluate GPT-4 (8K) on a subset of multi-document QA experiments, finding similar results and trends as other models (though GPT-4 has higher absolute performance). Evaluating GPT-4 on the full multi-document QA and key-value retrieval experiments would cost upwards of $6000. See Appendix D for GPT-4 results and discussion.

Input Context

```
Extract the value corresponding to the specified key in the JSON object below.

JSON data:
{"2a8d601d-1d69-4e64-9f90-8ad825a74195": "bb3ba2a5-7de8-434b-a86e-a88bb9fa7289",
 "a54e2eed-e625-4570-9f74-3624e77d6684": "d1ff29be-4e2a-4208-a182-0cea716be3d4",
 "9f4a92b9-5f69-4725-ba1e-403f08dea695": "703a7ce5-f17f-4e6d-b895-5836ba5ec71c",
 "52a9c80c-da51-4fc9-bf70-4a4901bc2ac3": "b2f8ea3d-4b1b-49e0-a141-b9823991ebeb",
 "f4eb1c53-af0a-4dc4-a3a5-c2d50851a178": "d733b0d2-6af3-44e1-8592-e5637fdb76fb"}

Key: "9f4a92b9-5f69-4725-ba1e-403f08dea695"
Corresponding value:
```

Desired Output

```
703a7ce5-f17f-4e6d-b895-5836ba5ec71c
```

Figure 6: Example of the key-value retrieval task, with an input context and the desired model output. Given a key, the goal is to return the associated value. All keys and values are 128-bit UUIDs. The relevant key-value pair for answering the query is bolded here within the input context for clarity.

indicate that extended-context models are not necessarily better than their non-extended counterparts at using their input context.

3 How Well Can Language Models Retrieve From Input Contexts?

Given that language models struggle to retrieve and use information from the middle of their input contexts in the multi-document question answering task, to what extent can they simply *retrieve* from input contexts? We study this question with a synthetic key-value retrieval task, which is designed to provide a minimal testbed for the basic ability to retrieve matching tokens from an input context.

3.1 Experimental Setup

In our synthetic key-value retrieval task, the inputs are (i) a string-serialized JSON object with k key-value pairs, where each of the keys and values are unique, randomly-generated UUIDs and (ii) a key within the aforementioned JSON object. The goal is to return the value associated with the specified key. Thus, each JSON object contains one relevant key-value pair (where the value is to be returned), and $k - 1$ irrelevant "distractor" key-value pairs. Figure 6 provides an example input context and its corresponding desired output. We again measure accuracy by evaluating whether the correct value appears in the predicted output.

Our synthetic key-value retrieval task shares similar goals with the Little Retrieval Test of Papailiopoulos et al. (2023) and the fine-grained line retrieval task of Li et al. (2023), but we explicitly seek to distill and simplify the task by removing as much natural language semantics as possible (using random UUIDs instead), since language features may present potential confounders. For example, Transformer language models may have varying sensitivity to different linguistic features in their input (O'Connor and Andreas, 2021).

To modulate the position of relevant information within the input context, we change the position of the key to retrieve within the serialized JSON object. To modulate the input context length, we change the number of input JSON key-value pairs k by adding or removing random keys, changing the number of distractor key-value pairs.

3.2 Results and Discussion

We experiment with input contexts containing 75, 140, and 300 key-value pairs (500 examples each). We use the same set of models as the multi-document question answering experiments, see §2.2 for more details.

Figure 7 presents key-value retrieval performance. Claude-1.3 and Claude-1.3 (100K) do nearly perfectly on all evaluated input context lengths, but other models struggle, especially when contexts have 140 or 300 key-value pairs—although the synthetic key-value retrieval task only requires identifying exact match within the input context, not all models achieve high performance.

Similar to our multi-document QA results, GPT-3.5-Turbo, GPT-3.5-Turbo (16K), and MPT-30B-Instruct have the lowest performance when they must access key-value pairs in the middle of their input context. LongChat-13B (16K) exhibits a different trend in the 140 key-value setting; we qualitatively observe that when relevant information is